NEURALPATH CHEATSHEET

Module 2: Classical Machine Learning

Master regression, gradient descent, classification, regularization, tree ensembles, SVMs, clus

Core Formulas, Mental Models & Engineering Rules

- Always split train/validation/test BEFORE fitting scalers, encoders, or PCA (no leakage).
- Gradient Descent: theta <- theta - eta * grad(L). Scale features to fix ravine conditioning.
- Bias-Variance: Ridge (L2) shrinks weights smoothly; Lasso (L1) drives sparse weights to zero.
- Metrics: use PR-AUC and F1 under severe class imbalance; inspect calibration and confusion matrix.
- Ensembles: Random Forests reduce variance via bagging; Gradient Boosting reduces bias sequentially.

Estimated study time: 6 hours · Badge: Classical Practitioner